

The Network Design Studio

Six problems. You choose the network. The choice is the answer.

Companion to *Where the Loops Come From* • MYP

How to draw. Join dots with straight lines only. On the **square** grid a sideways or upright step is **1 unit** and a diagonal is **1.4 units**. On the **triangular** grid every step to a touching dot is **1 unit**, which makes lengths easy to add up and is why some of these problems use it.

How to score. *Length* is the total of all your steps. *Loops* is the number of separate rings in your network. To find *links whose loss cuts someone off*, rub out each link one at a time and ask whether every important dot can still be reached.

1 The village water supply

A reservoir sits at the top of the hill (**R**). Five houses sit below it. Water only ever flows downhill, from R to the houses, and the pipes are new and will not break. Pipe costs money by the unit.

Draw the cheapest network that reaches all five houses. Mark R and the houses first.



Score it	length (units)	loops	links whose loss cuts someone off
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1a. How many loops did you draw? Explain why, in one sentence, using the word *downhill*.

2 The same village, one hard winter

Same reservoir, same five houses, same positions. But now any single pipe can freeze and burst. You are given a budget for exactly **two extra links** beyond your answer to problem 1.

Redraw the network and spend the two extra links. You must choose *where*.



Score it	length (units)	loops	links whose loss cuts someone off
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2a. Which house was in the most danger before you spent the extra links, and how did you know?

2b. Your network is now longer than in problem 1. Say exactly what you bought with that extra length.

3 The slime mould problem

Six food dots, marked **F**, scattered on the triangular grid. Anything may need to travel from any food dot to any other, in either direction, at any time. One link will be cut by partner at the end, and you will not be told which until after you have drawn.

Draw a network joining all six. Your total length must not exceed 14 units. Place six F dots yourself, spread out.



Score it	length (units)	loops	links whose loss cuts someone off
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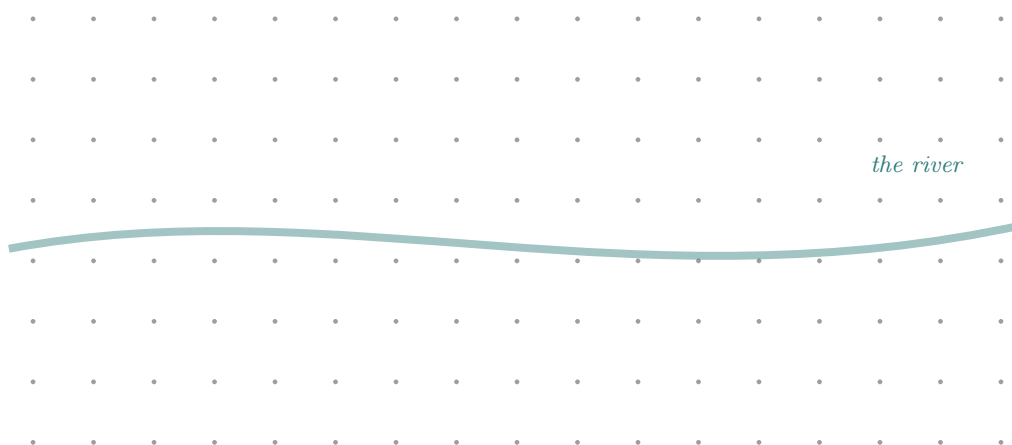
3a. After the cut: is every F still reachable? YES NO

3b. If you had been told in advance which link would be cut, would you have drawn a different network? Explain what that tells you about why loops exist.

4 Two bridges or one

A river runs across the grid. Ordinary links cost **1 unit**. A bridge across the river costs **6 units**. There are houses on both banks, and everybody travels to everybody.

Decide: one bridge or two? Draw your answer, then justify the cost.



Score it	length (units)	loops	links whose loss cuts someone off
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4a. Total cost of your answer: _____ units. Cost of the other choice: _____ units.

4b. Under what circumstances would the second bridge be worth it, and under what circumstances would it be a waste? Give one of each.

5 The fungus: draw before you know

A fungus spreads through soil *before* it knows where the food is. Draw your network on the grid below now. Then your teacher will roll for four food positions, which you will mark afterwards.

Your network may use at most 20 units. You are trying to be close to wherever the food turns out to be.



Score it	length used	loops	steps from your network to the 4 food dots
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5a. Compare with two people near you. Did the person with the *shortest* network do best? What did the best network have in common?

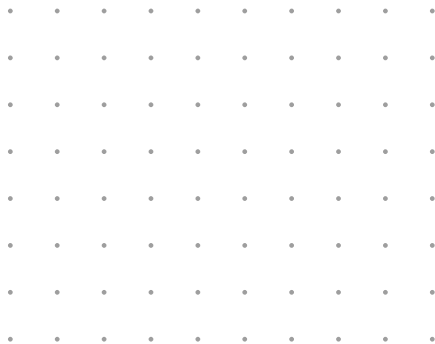
5b. A real fungus does not have to commit in advance; it grows, then withers the parts that were useless. What advantage does that give it over your single drawn answer?

6 Now choose the grid yourself

One last problem, and this time the choice of *grid* is part of the answer. You are designing streets for a small new town on flat ground, with eight destinations, and traffic that changes hour by hour.

Circle the grid you will use, then draw on it. Square, or triangular?

SQUARE



TRIANGULAR



6a. Why did you choose that grid? Say something about the directions it lets you travel in.

6b. Real cities are mostly built on something closer to the square grid, yet honeycombs and many natural networks meet at angles closer to the triangular one. Suggest one reason people build differently from bees.

Teacher key and notes

Problem 1. Expect a tree, and expect nobody to draw a loop unprompted. If a student does, that is worth surfacing: ask what the second route into a house would ever be used for, given water only comes from R. Length should land near the minimum spanning tree of whatever house positions they chose; exact optimality is not the point.

Problem 2. The productive difficulty is that two extra links cannot protect everything, so the student must decide which failure matters most. Strong answers protect the link nearest R, since its loss disconnects the most houses; this is the idea of a *bridge* in graph theory arriving on its own. Weak answers add links between two leaves far from R, which protects almost nothing. Both are worth comparing on the board.

Problem 3. Run the cut publicly: pick a link at random from each student's drawing rather than choosing one, or the exercise becomes a game against you rather than against uncertainty. Expect roughly half the class to be disconnected on the first attempt. Question 3b is the hinge of the whole unit: knowing which link fails removes the need for the loop entirely, which is precisely why predictable networks are trees.

Problem 4. The arithmetic is deliberately close. One bridge is cheaper and is correct if the bridge never closes; two is correct if it might. The intended discovery is that the answer depends on a fact about the world that is not on the diagram, and students who say "it depends" and then say on what have given the best possible answer.

Problem 5. Use six-sided dice for grid coordinates, or draw four positions from a hat prepared in advance. The winning networks are usually spread and moderately loopy rather than short and compact, which is worth naming: this is the same trade-off as problem 3 but with the uncertainty in *where*, not in *which link fails*. Question 5b points at the real difference between a designed network and an adaptive one.

Problem 6. There is no correct grid. The triangular grid gives six directions and equal step lengths, so routes are more direct and every link costs the same; the square grid gives four directions and awkward diagonals, but it packs rectangular buildings and plots without waste. That last point is the intended answer to 6b: people build square because rooms, fields and bricks are square, and the street network inherits the shape of the things beside it rather than the shape that would be best for travelling.

Timing. Problems 1–3 are one lesson with the discussion; 4–6 are a second. If only one lesson is available, do 1, 2 and 3 and set 4 as homework, since 3 without 1 and 2 loses the contrast it depends on.